第三章

角动量及对称性理论

§1 谐角动量算符

1.1 角动量算符的定义 （广义）

$\hat{J}(\hat{J}_x, \hat{J}_y, \hat{J}_z)$ 矢量算符（厄米）

定义式：

$$\boxed{\hat{J} \times \hat{J} = i\hbar \hat{J}} \qquad (\hbar = 1)$$

分量式：

$$[\hat{J}_x, \hat{J}_y] = i\hat{J}_z,$$

$$[\hat{J}_y, \hat{J}_z] = i\hat{J}_x,$$

$$[\hat{J}_z, \hat{J}_x] = i\hat{J}_y.$$

（x·y·z 循环）

概括描述

$$\Longrightarrow \boxed{[\hat{J}_i, \hat{J}_j] = i\sum_k \epsilon_{ijk}\hat{J}_k}$$

$i, j, k = x, y, z$

Levi-Civita 符号 ϵ_{ijk}

$$\epsilon_{ijk} = \begin{cases} 1 & ijk = xyz, yzx, zxy \\ -1 & ijk = xzy, yxz, zyx \\ 0 & \text{其它情况}. \end{cases}$$

例：量子系统中的角动量

① $\hat{J} = \hat{L} = \hat{r} \times \hat{p}$ （轨道角动量）

② $\hat{J} = \hat{S}$ （自旋角动量）

③ $\hat{J} = \hat{L} + \hat{S}$ （总角动量）

1.2 角动量平方算符.

$$\hat{J}^2 = \hat{J} \cdot \hat{J} = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2$$

易证：

$$[\hat{J}^2, \hat{J}_i] = 0 \quad (i = x, y, z)$$

1.3 上升算符和下降算符

定义：

上升算符：$\boxed{\hat{J}_+ \equiv \hat{J}_x + i\hat{J}_y}$

下降算符：$\boxed{\hat{J}_- \equiv \hat{J}_x - i\hat{J}_y}$

- $\hat{J}_+, \hat{J}_-$ 都不是厄密算符.

$$(\hat{J}_+)^\dagger = \hat{J}_- \quad , \quad (\hat{J}_-)^\dagger = \hat{J}_+$$

互为厄密共轭！

$$\boxed{[\hat{J}^2, \hat{J}_\pm] = 0}$$

- $\hat{J}_+\hat{J}_-$，$\hat{J}_-\hat{J}_+$ 为厄密算符

- $\hat{J}_z$, $\hat{J}_+$, $\hat{J}_-$ 间的对易关系

$$\boxed{[\hat{J}_z, \hat{J}_+] = \hat{J}_+},$$

$$\boxed{[\hat{J}_z, \hat{J}_-] = -\hat{J}_-},$$

$$\boxed{[\hat{J}_+, \hat{J}_-] = 2\hat{J}_z}$$

- 逆变换

$$\boxed{\hat{J}_x = \frac{\hat{J}_+ + \hat{J}_-}{2}}$$

$$\boxed{\hat{J}_y = \frac{\hat{J}_+ - \hat{J}_-}{2i}}$$

$$\cdot\ \hat{J}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2$$

$$= \frac{1}{2}\left[\hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+\right] + \hat{J}_z^2$$

利用 $[\hat{J}_+, \hat{J}_-] = 2\hat{J}_z$

$$\Rightarrow \quad (a)\quad \hat{J}^2 = \hat{J}_+\hat{J}_- - \hat{J}_z + \hat{J}_z^2$$

$$(b)\quad \hat{J}^2 = \hat{J}_-\hat{J}_+ + \hat{J}_z + \hat{J}_z^2$$

$$\Rightarrow \quad (a')\quad \hat{J}_+\hat{J}_- = \hat{J}^2 - \hat{J}_z(\hat{J}_z - 1)$$

$$(b')\quad \hat{J}_-\hat{J}_+ = \hat{J}^2 - \hat{J}_z(\hat{J}_z + 1)$$

§2 角动量算符的本征值与本征矢

2.1 $(\hat{J}^2, \hat{J}_z)$-表象

$$[\hat{J}^2, \hat{J}_i] = 0 \quad .$$

共同本征矢： $|\lambda, m\rangle$

$$\begin{cases} \hat{J}^2 |\lambda, m\rangle = \lambda |\lambda, m\rangle , \\ \hat{J}_z |\lambda, m\rangle = m |\lambda, m\rangle , \end{cases}$$

(λ, m 通常为实量)

- $\langle \lambda', m' | \lambda, m \rangle = \delta_{\lambda'\lambda} \delta_{m'm}$

- $\sum_{\lambda} \sum_{m} |\lambda m\rangle\langle \lambda m| = \mathbb{1}$.

2.2 两个不等式

(1) $\lambda \geq 0$ → $\hat{J}\cdot\hat{J} = \hat{J}^{\dagger}\cdot\hat{J}$

证明：$\lambda = \langle \lambda, m | \hat{J}^2 | \lambda, m \rangle$

$$= \sum_{i=x,y,z} \langle \lambda, m | \hat{J}_i^{\dagger} \hat{J}_i | \lambda, m \rangle$$

$$= \sum_{i=x,y,z} | \hat{J}_i | \lambda, m \rangle |^2 \geq 0$$

证毕.

(2) $\lambda \geq m^2$

证明：$\lambda = \sum_{i=x,y} \langle \lambda m | \hat{J}_i^{\dagger} \hat{J}_i | \lambda m \rangle$

$$+ \langle \lambda, m | \hat{J}_z^{\dagger} \hat{J}_z | \lambda m \rangle$$

$$= \sum_{i=x,y} | \hat{J}_i | \lambda, m \rangle |^2 + m^2$$

$\Rightarrow \lambda - m^2 \geq 0$ 即 $\lambda \geq m^2$ 证毕.

2.3 其它本征态

(a) $|\psi\rangle \equiv \hat{J}_+ |\lambda, m\rangle$ 是否为 $\hat{J}^2$ 与 $\hat{J}_z$ 的共同本征态？

- $\hat{J}^2|\psi\rangle = \hat{J}^2 \hat{J}_+ |\lambda, m\rangle = \hat{J}_+ \hat{J}^2 |\lambda, m\rangle$

$= \lambda \hat{J}_+ |\lambda m\rangle = \lambda |\psi\rangle$

$\Rightarrow$ $\hat{J}_+^p |\lambda, m\rangle$ 是 $\hat{J}^2$ 的简并本征态！

(p：正整数)

- $\hat{J}_z |\psi\rangle = \hat{J}_z \hat{J}_+ |\lambda, m\rangle$

$= \left([\hat{J}_z, \hat{J}_+] + \hat{J}_+ \hat{J}_z \right) |\lambda, m\rangle$

$= \left(\hat{J}_+ + \hat{J}_+ \hat{J}_z \right) |\lambda, m\rangle$

$= \left(\hat{J}_+ + m \hat{J}_+ \right) |\lambda, m\rangle$

$= (1 + m) \hat{J}_+ |\lambda, m\rangle$

$= (1+m) |\psi\rangle$

$\Rightarrow$ $\hat{J}_+|\lambda,m\rangle$ 是 $\hat{J}_z$ 的本征值为 $(m+1)$ 的本征态.

$$\hat{J}_+|\lambda,m\rangle \propto |\lambda,m+1\rangle$$

$\Rightarrow$ $\hat{J}_+$：上升算符

(b) 类似可得：

$$\hat{J}_z(\hat{J}_+)^p|\lambda,m\rangle = (m+p)(\hat{J}_+)^p|\lambda m\rangle$$

$$(\hat{J}_+)^p|\lambda,m\rangle \propto |\lambda,m+p\rangle$$

p：正整数

(c) $|\varphi\rangle \equiv \hat{J}_-|\lambda,m\rangle$ 是否为 $\hat{J}^2$ 与 $\hat{J}_z$ 的共同本征态？

- $\hat{J}^2|\varphi\rangle = \hat{J}^2\hat{J}_-|\lambda,m\rangle = \hat{J}_-\hat{J}^2|\lambda,m\rangle$

$$= \lambda \hat{J}_-|\lambda,m\rangle = \lambda|\varphi\rangle$$

$\Rightarrow$ $\hat{J}_-^p|\lambda,m\rangle$ 是 $\hat{J}^2$ 的简并本征态！ (p：正整数)

- $\hat{J}_z|\varphi\rangle = \hat{J}_z\hat{J}_-|\lambda,m\rangle$

$$= \left([\hat{J}_z,\hat{J}_-] + \hat{J}_-\hat{J}_z\right)|\lambda,m\rangle$$

$$= \left(-\hat{J}_- + \hat{J}_- m\right)|\lambda,m\rangle$$

$$= (m-1)\hat{J}_-|\lambda,m\rangle$$

$$= (m-1)|\varphi\rangle$$

$\Rightarrow$ $\hat{J}_-|\lambda, m\rangle$ 是 $\hat{J}_z$ 的本征值为 $(m-1)$ 的本征态.

$$\hat{J}_-|\lambda, m\rangle \propto |\lambda, m-1\rangle$$

$\Rightarrow$ $\hat{J}_-$：下降算符.

(d) 类似得：

$$\hat{J}_z\ (\hat{J}_-)^q|\lambda, m\rangle = (m-q)\overbrace{(\hat{J}_-)^q}|\lambda, m\rangle$$

$$(\hat{J}_-)^q|\lambda, m\rangle \propto |\lambda, m-q\rangle$$

q：正整数

- 确定比例系数

(1) $\hat{J}_+ |\lambda\ m\rangle \propto |\lambda, m+1\rangle$

令 $C_m \hat{J}_+ |\lambda m\rangle \equiv |\lambda, m+1\rangle$，确定 C_m.

要求：$\langle \lambda, m+1 | \lambda, m+1\rangle = 1$，

但，$\langle \lambda, m+1| = \langle \lambda m | C_m^* (\hat{J}_+)^\dagger$

$= \langle \lambda\ m | C_m^* \hat{J}_-$，

$\Rightarrow \quad \langle \lambda, m+1 | \lambda, m+1\rangle$

$= |C_m|^2 \langle \lambda m | \hat{J}_- \hat{J}_+ | \lambda m\rangle$

$= |C_m|^2 \langle \lambda m | [\hat{J}_-^2 - \hat{J}_z(\hat{J}_z+1)] | \lambda, m\rangle$

$= |C_m|^2 (\lambda - m(m+1)) = 1$

$\Rightarrow \quad |C_m|^2 = \dfrac{1}{\lambda - m(m+1)}$

$\Rightarrow \quad \dfrac{1}{|C_m|^2} = \lambda - m(m+1)$

· 取 C_m 为实数： $\frac{1}{C_m} = \sqrt{\lambda - m(m+1)}$

· 再令 $\lambda \equiv j(j+1)$ ，则 $|\lambda\ m\rangle \to |j, m\rangle$ ，

$$\frac{1}{C_m} = \sqrt{j(j+1) - m(m+1)} = \sqrt{(j-m)(j+m+1)}$$

$$\Rightarrow \boxed{\hat{J}_+ |j, m\rangle = \sqrt{(j-m)(j+m+1)}\, |j\ m+1\rangle} \quad \cdots(1)$$

↳ $\hat{J}_+ |j, j\rangle = 0$

(2) $\hat{J}_- |j, m\rangle \propto |j, m-1\rangle$

令 $C_m' \hat{J}_- |j\ m\rangle \equiv |j, m-1\rangle$.

类似可得： $\frac{1}{C_m'} = \sqrt{(j+m)(j-m+1)}$

$$\Rightarrow \boxed{\hat{J}_- |j\ m\rangle = \sqrt{(j+m)(j-m+1)}\, |j, m-1\rangle} \quad \cdots(2)$$

↳ $\hat{J}_- |j, -j\rangle = 0$

· $\lambda \geq m^2 \Rightarrow j(j+1) \geqslant m^2$

2.4 j 与 m 的可取值

※ Schwinger 的方法

- 引入两套互对易的玻色型产生算符和湮灭算符：

$$\hat{a}_1, \hat{a}_1^\dagger \quad ; \quad \hat{a}_2, \hat{a}_2^\dagger \quad \text{(非厄米算符)}$$

- 对易关系：

$$[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}, \quad i,j = 1,2$$

$$[\hat{a}_i, \hat{a}_j] = 0,$$

$$[\hat{a}_i^\dagger, \hat{a}_j^\dagger] = 0,$$

- 引入真空态：$|0\rangle$ （$\langle 0|0\rangle = 1$）

$$\hat{a}_i|0\rangle = 0 \quad , \quad i=1,2$$

- 粒子数算符：$\hat{N}_1$、$\hat{N}_2$

$$\hat{N}_i \equiv \hat{a}_i^\dagger \hat{a}_i \quad , \ i=1,2 \quad \text{（厄密算符）}$$

- 本征值方程

$$\begin{cases} \hat{N}_i|n_i\rangle = n_i|n_i\rangle \quad , \ i=1,2 \\ n_i = 0, 1, 2, \cdots \\ |n_i\rangle = \frac{1}{\sqrt{n_i!}}(\hat{a}_i^\dagger)^{n_i}|0\rangle \quad , \ i=1,2 \end{cases}$$

- 单i子系统基矢：$\{|n_i\rangle\}$，

$\langle n_i|n_i'\rangle = \delta_{nn'}$，$\sum_{n_i}|n_i\rangle\langle n_i| = \mathbb{1}$.

- $[\hat{N}_1, \hat{N}_2] = 0$ ；共同本征态

$$|n_1, n_2\rangle = |n_1\rangle \otimes |n_2\rangle \quad \text{（复合系统基矢）}$$

$$= \frac{1}{\sqrt{n_1!}} (\hat{a}_1^\dagger)^{n_1} \otimes \frac{1}{\sqrt{n_2!}} (\hat{a}_2^\dagger)^{n_2} |0\rangle$$

$$= \frac{1}{\sqrt{n_1! n_2!}} (\hat{a}_1^\dagger)^{n_1} (\hat{a}_2^\dagger)^{n_2} |0\rangle$$

$\left(\hat{N}_1 |n_1, n_2\rangle = n_1 |n_1, n_2\rangle, \ \hat{N}_2 |n_1, n_2\rangle = n_2 |n_1, n_2\rangle \right)$

- 引入角动量算符：$\hat{J}$

$$\hat{J}_l \equiv \left(\hat{a}_1^\dagger, \hat{a}_2^\dagger \right) \frac{\hat{\sigma}_l}{2} \begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \end{pmatrix} \quad (l = x, y, z),$$

其中 $\hat{\sigma}(\sigma_x, \sigma_y, \sigma_z)$ 为 Pauli 矩阵：

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

$$\Rightarrow \begin{cases} \hat{J}_x = \frac{1}{2}\left(\hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1\right) = \hat{J}_x^\dagger, \\ \hat{J}_y = \frac{1}{2i}\left(\hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1\right) = \hat{J}_y^\dagger, \\ \hat{J}_z = \frac{1}{2}\left(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2\right) \\ \quad = \frac{1}{2}\left(\hat{N}_1 - \hat{N}_2\right) = \hat{J}_z^\dagger \end{cases}$$

厄密

又，
$$\begin{cases} \hat{J}_+ \equiv \hat{J}_x + i\hat{J}_y = \hat{a}_1^\dagger \hat{a}_2 \\ \hat{J}_- \equiv \hat{J}_x - i\hat{J}_y = \hat{a}_2^\dagger \hat{a}_1 \end{cases}$$

非厄密
$(\hat{J}_+)^\dagger = \hat{J}_-$
$(\hat{J}^\dagger \hat{J}^-$：厄密)

$$\Rightarrow \hat{J}^2 \equiv \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2 = \frac{\hat{N}}{2}\left(\frac{\hat{N}}{2}+1\right),$$

其中，$\hat{N} \equiv \hat{N}_1 + \hat{N}_2$

为总粒子数算符．　$\hat{N}^\dagger = \hat{N}$（厄密算符）

증명: $\hat{J}^2 = \frac{\hat{N}}{2}\left(\frac{\hat{N}}{2}+1\right)$.

$$\hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2 = \frac{1}{4}\left[\hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1\right]^2$$

$$- \frac{1}{4}\left[\hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1\right]^2$$

$$+ \frac{1}{4}\left[\hat{N}_1 - \hat{N}_2\right]^2$$

$$= \frac{1}{2}\left[\hat{a}_1^\dagger \hat{a}_2 \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_1 \hat{a}_1^\dagger \hat{a}_2\right]$$

$$+ \frac{1}{4}\left[\hat{N}_1^2 + \hat{N}_2^2 - 2\hat{N}_1\hat{N}_2\right]$$

$$= \frac{1}{2}\left[\hat{a}_1^\dagger \hat{a}_1 (1 + \hat{a}_2^\dagger \hat{a}_2) + \hat{a}_2^\dagger \hat{a}_2 (1 + \hat{a}_1^\dagger \hat{a}_1)\right]$$

$$+ \frac{1}{4}\left[\hat{N}_1^2 + \hat{N}_2^2 - 2\hat{N}_1\hat{N}_2\right]$$

$$= \frac{1}{2}\left[\hat{N}_1 (1 + \hat{N}_2) + \hat{N}_2 (1 + \hat{N}_1)\right]$$

$$+ \frac{1}{4}\left[\hat{N}_1^2 + \hat{N}_2^2 - 2\hat{N}_1\hat{N}_2\right]$$

$$= \frac{1}{2}(\hat{N}_1 + \hat{N}_2) + \frac{1}{4}(\hat{N}_1^2 + \hat{N}_2^2 + 2\hat{N}_1\hat{N}_2)$$

$$= \frac{\hat{N}}{2} + \frac{1}{4}\hat{N}^2 = \frac{\hat{N}}{2}\left(\frac{\hat{N}}{2}+1\right)$$

- 可证，引入的角动算符满足 希-空间 诸角动间的所有对易关系.

- $\{\hat{J}_z, \hat{J}^2, \hat{N}_1, \hat{N}_2\}$ 互对易！

⇒ $|n_1, n_2\rangle$ 亦为 $\hat{J}_z$ 和 $\hat{J}^2$ 算符的本征态.

$$\hat{J}_z |n_1, n_2\rangle = \frac{1}{2}(\hat{N}_1 - \hat{N}_2)|n_1, n_2\rangle$$

$$= \frac{1}{2}(n_1 - n_2)|n_1, n_2\rangle$$

$$\hat{J}^2 |n_1, n_2\rangle = \frac{\hat{N}}{2}\left(\frac{\hat{N}}{2}+1\right)|n_1, n_2\rangle$$

$$= \left(\frac{n_1+n_2}{2}\right)\left[\frac{(n_1+n_2)}{2}+1\right]|n_1, n_2\rangle$$

$|n_1, n_2\rangle$：粒子数表象中 $\hat{J}_z$ 和 $\hat{J}^2$ 算符的共同本征态！

- $|jm\rangle \Rightarrow |n_1, n_2\rangle = \frac{(\hat{a}_1^\dagger)^{n_1}}{\sqrt{n_1!}} \frac{(\hat{a}_2^\dagger)^{n_2}}{\sqrt{n_2!}} |0\rangle$

- 本程序.

$$\begin{cases} \hat{J}_z |jm\rangle = m |jm\rangle \\ \hat{J}_z |n_1 n_2\rangle = \frac{1}{2}(n_1 - n_2) |n_1 n_2\rangle \end{cases}$$

$$\Rightarrow \quad \boxed{m = \frac{1}{2}(n_1 - n_2)} \qquad \cdots (1)$$

$$\begin{cases} \hat{J}^2 |jm\rangle = j(j+1) |jm\rangle \\ \hat{J}^2 |n_1 n_2\rangle = \left(\frac{n_1+n_2}{2}\right)\left[\left(\frac{n_1+n_2}{2}\right)+1\right] |n_1 n_2\rangle \end{cases}$$

$$\Rightarrow \quad \boxed{j(j+1) = \left(\frac{n_1+n_2}{2}\right)\left(\frac{n_1+n_2}{2}+1\right)} \quad \cdots (2)$$

$$\Rightarrow \quad \boxed{j = \frac{n_1+n_2}{2} = \begin{cases} 0, 1, 2 \cdots & (n_1+n_2 \text{ 为偶数}) \\ \frac{1}{2}, \frac{3}{2}, \frac{5}{2} \cdots & (n_1+n_2 \text{ 为奇数}) \end{cases}} \quad \cdots (3)$$

即 $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \cdots$ $\cdots$ (A)

$$\boxed{m = \frac{1}{2}(n_1-n_2) = \begin{cases} 0, \pm 1, \pm 2 \cdots & (n_1-n_2 \text{ 为偶数}) \\ \pm\frac{1}{2}, \pm\frac{3}{2}, \pm\frac{5}{2} \cdots & (n_1-n_2 \text{ 为奇数}) \end{cases}} \quad \cdots (4)$$

即 $m = 0, \pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}, \cdots$ $\cdots$ (B)

由 $j = \frac{n_1+n_2}{2}$, $m = \frac{n_1-n_2}{2}$ 得

$n_1 = j+m$, $\cdots$ (a)

$n_2 = j-m$. $\cdots$ (b)

$m^2 \leq j(j+1)$

$$|j\ m\rangle = |n_1, n_2\rangle$$

$$= \frac{(\hat{a}_1^\dagger)^{n_1}}{\sqrt{n_1!}} \frac{(\hat{a}_2^\dagger)^{n_2}}{\sqrt{n_2!}} |0\rangle$$

将 $n_1 = j+m$

$n_2 = j-m$ 代入上式，得

$$\boxed{|j\ m\rangle = \frac{(\hat{a}_1^\dagger)^{j+m}}{\sqrt{(j+m)!}} \frac{(\hat{a}_2^\dagger)^{j-m}}{\sqrt{(j-m)!}} |0\rangle .}$$

例 $j=1$ 情况.

$m_1^2 \le j(j+1) \Rightarrow m^2 \le 2$

(a) $n_1 = l + m$

- $n_1 = 0$, $m = -1$, $m^2 = 1$ ✓
- $n_1 = 1$, $m = 0$, $m^2 = 0$ ✓
- $n_1 = 2$, $m = 1$, $m^2 = 1$ ✓
- $n_1 = 3$, $m = 2$, $m^2 = 4$ ✗

⋮

(b) $n_2 = l - m$

- $n_2 = 0$, $m = 1$, $m^2 = 1$ ✓
- $n_2 = 1$, $m = 0$, $m^2 = 0$ ✓
- $n_2 = 2$, $m = -1$, $m^2 = 1$ ✓
- $n_2 = 3$, $m = -2$, $m^2 = 4$ ✗

⋮

$m = -1, 0, +1$

一般情况：

$$m = -j, -j+1, \cdots j-1, j$$

$2j+1$ ↑

- $|j, m=j\rangle = \dfrac{(\hat{a}_1^\dagger)^{2j}}{\sqrt{(2j)!}}|0\rangle$

- $|j, m=-j\rangle = \dfrac{(\hat{a}_2^\dagger)^{2j}}{\sqrt{(2j)!}}|0\rangle$

(a) $\hat{J}_+ |j, j\rangle$

$= \hat{a}_1^\dagger \hat{a}_2 \dfrac{(\hat{a}_1^\dagger)^{2j}}{\sqrt{(2j)!}}|0\rangle$

$= \hat{a}_1^\dagger \dfrac{(\hat{a}_1^\dagger)^{2j}}{\sqrt{(2j)!}} \hat{a}_2|0\rangle = 0$

(b) $J_- |j, -j\rangle$

$= \hat{a}_2^\dagger \hat{a}_1 \dfrac{(\hat{a}_2^\dagger)^{2j}}{\sqrt{(2j)!}}|0\rangle$

$= \hat{a}_2^\dagger \dfrac{(\hat{a}_2^\dagger)^{2j}}{\sqrt{(2j)!}} \hat{a}_1|0\rangle = 0$

· 公式

(a) $$|j, \pm m\rangle = \left[\frac{(j+m)!}{(2j)!\,(j-m)!} \right]^{\frac{1}{2}} (\hat{J}_-)^{j-m} |j, \pm j\rangle .$$

(b) $$|j, \pm j\rangle = \left[\frac{(j+m)!}{(2j)!\,(j-m)!} \right]^{\frac{1}{2}} (\hat{J}_+)^{j-m} |j, \pm m\rangle .$$

§3 角动量算符的矩阵元.

3.1 $\{|jm\rangle\}$ 的正交、完备性

$$\begin{cases} \langle j'm'|jm\rangle = \delta_{jj'}\delta_{m'm} \\ \sum_j \sum_m |jm\rangle\langle jm| = \mathbb{1} \end{cases}$$

· 对给定的 j 值. (子空间)

$$\begin{cases} \langle jm'|jm\rangle = \delta_{m'm}. \\ \sum_{m=-j}^{j} |jm\rangle\langle jm| = \mathbb{1}. \end{cases}$$

$(2j+1)$ 维希尔伯特空间

3.2 诺特动量算符的矩阵元

$$\hat{J}' \equiv \hat{J}_x, \hat{J}_y, \hat{J}_z, \hat{J}_+, \hat{J}_-, \hat{J}^2$$

$$[\hat{J}', \hat{J}^2] = 0$$

$$\Rightarrow 0 = \langle j'm' | [\hat{J}', \hat{J}^2] | jm \rangle$$

$$= \langle j'm' | (\hat{J}'\hat{J}^2 - \hat{J}^2\hat{J}') | jm \rangle$$

$$= [j(j+1) - j'(j'+1)] \langle j'm' | \hat{J}' | jm \rangle$$

若 $j' \neq j$ 则 $\langle j'm' | \hat{J}' | jm \rangle = 0$

即 $\langle j'm' | \hat{J}' | jm \rangle \propto \delta_{jj'}$

关于 j，只有对角元不恒为零！

· 对 $j'=j$ 的子空间

(a) $\langle j m' | \hat{J}^2 | j m \rangle = j(j+1)\,\delta_{m'm}$

(b) $\langle j m' | \hat{J}_z | j m \rangle = m\,\delta_{m'm}$

(c) $\langle j m' | \hat{J}_\pm | j m \rangle = \delta_{m',m\pm1} \times \sqrt{(j \mp m)(j \pm m + 1)}$

(d) $\langle j m' | \hat{J}_x | j m \rangle = \langle j m' | \left(\frac{\hat{J}_+ + \hat{J}_-}{2} \right) | j m \rangle$

(e) $\langle j m' | \hat{J}_y | j m \rangle = \langle j m' | \left(\frac{\hat{J}_+ - \hat{J}_-}{2i} \right) | j m \rangle$

例1: $j=\frac{1}{2}$，求 $\hat{J}_x, \hat{J}_y, \hat{J}_z$ 的矩阵表示。 $\underbrace{\langle j=\frac{1}{2}, m' | \hat{J}_i | j=\frac{1}{2}, m \rangle}_{\text{矩阵元}}$

$\begin{cases} j=\frac{1}{2}, \\ m'=\frac{1}{2}, -\frac{1}{2} \\ m=\frac{1}{2}, -\frac{1}{2} \end{cases}$

m' \ m	$\frac{1}{2}$	$-\frac{1}{2}$
$\frac{1}{2}$		
$-\frac{1}{2}$		

可得：

$$J_x = \frac{1}{2}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$J_y = \frac{1}{2}\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix},$$

$$J_z = \frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

定义 $J_i = \frac{1}{2}\sigma_i$ （Pauli 矩阵）

fall 2 : $j = 1$ $\quad m', m = 1, 0, -1$

$$J_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$J_y = \frac{i}{\sqrt{2}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$J_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

§4 角动量的耦合

（两个角动量的耦合）

4.1 子系统角动量 $\hat{J}_1$ 和 $\hat{J}_2$

$$\hat{J}_1 \times \hat{J}_1 = i\hat{J}_1 \qquad \hat{J}_2 \times \hat{J}_2 = i\hat{J}_2$$

$$[\hat{J}_{1i}, \hat{J}_1^2] = 0 \qquad [\hat{J}_{2j}, \hat{J}_2^2] = 0$$

$i, j = x, y, z.$

- $[\hat{J}_{1i}, \hat{J}_{2j}] = 0$

相互独立！

§1.2 复合系统角动量

定义：$\hat{J} \equiv \hat{J}_1 + \hat{J}_2$. （总角动量算符）

- $\hat{J} = (\hat{J}_x, \hat{J}_y, \hat{J}_z)$

$= (\hat{J}_{1x}+\hat{J}_{2x}, \hat{J}_{1y}+\hat{J}_{2y}, \hat{J}_{1z}+\hat{J}_{2z})$.

可证：$\hat{J} \times \hat{J} = i\hat{J}$,

$[\hat{J}^2, \hat{J}_i] = 0 \quad (i = x, y, z)$

引入：$\hat{J}_+$, $\hat{J}_-$ （上升、下降算符）

- $\hat{J}_+ \equiv \hat{J}_x + i\hat{J}_y$, $\hat{J}_- \equiv \hat{J}_x - i\hat{J}_y$

$= \hat{J}_{1+} + \hat{J}_{2+}$ $= \hat{J}_{1-} + \hat{J}_{2-}$

可证：$\hat{J}_z$，$\hat{J}_\pm$，$\hat{J}^2$ 之间的

相互对易关系与子系统中的相同！

· 复合系统的角动量算符的矩表示

$$\begin{cases} \hat{J}^{(1,2)} = \hat{J}^{(1)} \otimes \hat{J}^{(2)} \\ \hat{J}^{(1)} = \hat{J}^{(1)} \otimes \mathbb{1}^{(2)} \\ \hat{J}^{(2)} = \mathbb{1}^{(1)} \otimes \hat{J}^{(2)} \end{cases}$$

4.3 复合系统的无耦合表象 （j_1，j_2 给定）

第一子系统基矢 $\{|j_1 m_1\rangle\}$，$(2j_1+1)$ 维

第二子系统基矢 $\{|j_2 m_2\rangle\}$，$(2j_2+1)$ 维

· 取复合系统基矢为：$\hat{J}_1^2$，$\hat{J}_{1z}$，$\hat{J}_2^2$，$\hat{J}_{2z}$ 的共同本征态

$$\{|j_1 m_1\rangle \otimes |j_2 m_2\rangle\}$$

$(2j_1+1)\times(2j_2+1)$ 维

将 $|j_1 m_1\rangle \otimes |j_2 m_2\rangle$ 简写为 $|j_1 m_1 j_2 m_2\rangle$

基矢满足的诸关系：

(a) $\hat{J}_1^2 \left| j_1\ m_1\ j_2\ m_2 \right\rangle = j_1(j_1+1) \left| j_1\ m_1\ j_2\ m_2 \right\rangle$

(b) $\hat{J}_2^2 \left| j_1\ m_1\ j_2\ m_2 \right\rangle = j_2(j_2+1) \left| j_1\ m_1\ j_2\ m_2 \right\rangle$

(c) $\hat{J}_{1z} \left| j_1\ m_1\ j_2\ m_2 \right\rangle = m_1 \left| j_1\ m_1\ j_2\ m_2 \right\rangle$

(d) $\hat{J}_{2z} \left| j_1\ m_1\ j_2\ m_2 \right\rangle = m_2 \left| j_1\ m_1\ j_2\ m_2 \right\rangle$

(e) $\hat{J}_{1\pm} \left| j_1\ m_1\ j_2\ m_2 \right\rangle = \sqrt{(j_1 \mp m_1)(j_1 \pm m_1 + 1)} \left| j_1\ m_1 \pm 1\ j_2\ m_2 \right\rangle$

(f) $\hat{J}_{2\pm} \left| j_1\ m_1\ j_2\ m_2 \right\rangle = \sqrt{(j_2 \mp m_2)(j_2 \pm m_2 + 1)} \left| j_1\ m_1\ j_2\ m_2 \pm 1 \right\rangle$

- 正交性：

$$\langle j_1\, m_1'\, j_2\, m_2' \mid j_1\, m_1\, j_2\, m_2 \rangle = \delta_{m_1' m_1}\, \delta_{m_2' m_2} .$$

- 完备性：

$$\sum_{m_1} \sum_{m_2} |j_1\, m_1\, j_2\, m_2\rangle\langle j_1\, m_1\, j_2\, m_2| = \mathbb{1} .$$

$$|\psi\rangle = \mathbb{1}\,|\psi\rangle$$

$$= \sum_{m_1} \sum_{m_2} |j_1\, m_1\, j_2\, m_2\rangle\langle j_1\, m_1\, j_2\, m_2|\,\psi\rangle .$$

4.4 耦合表象 (j_1, j_2 给定)

· 4个算符互对易：$\{\hat{J}^2, \hat{J}_z, \hat{J}_1^2, \hat{J}_2^2\}$

· 共同本征态：$\{|j_1\ j_2\ j\ m\rangle\}$

$(2j_1+1)$ $(2j_2+1)$ 维空间基矢

(a) $\hat{J}^2|\rangle = j(j+1)|\rangle$,

(b) $\hat{J}_z|\rangle = m|\rangle$,

(c) $\hat{J}_1^2|\rangle = j_1(j_1+1)|\rangle$.

(d) $\hat{J}_2^2|\rangle = j_2(j_2+1)|\rangle$.

· 正交归一性

$$\langle j_1 j_2 j' m' | j_1 j_2 j m \rangle = \delta_{j'j} \delta_{m'm}$$

· 完备性

$$\sum_j \sum_m |j_1 j_2 j m\rangle\langle j_1 j_2 j m| = \mathbb{1}.$$

· $|\psi\rangle = \mathbb{1}|\psi\rangle$

$$= \sum_j \sum_m |j_1 j_2 j m\rangle\langle j_1 j_2 j m|\psi\rangle.$$

采用了耦合表象！

例：耦合表象的应用.

设 $\hat{H} = \lambda \hat{J}_1 \cdot \hat{J}_2$

. 耦合表象基矢 $\{|j_1 j_2 j m\rangle\}$ 是 $\hat{H}$ 的本征态！

证明：$\hat{J}^2 = \hat{J} \cdot \hat{J} = (\hat{J}_1 + \hat{J}_2) \cdot (\hat{J}_1 + \hat{J}_2)$

$= \hat{J}_1^2 + \hat{J}_2^2 + 2\hat{J}_1 \cdot \hat{J}_2$

$\Rightarrow \quad \hat{J}_1 \cdot \hat{J}_2 = \dfrac{\hat{J}^2 - \hat{J}_1^2 - \hat{J}_2^2}{2}$.

$$\hat{H} |j_1 j_2 j m\rangle = \lambda \left[\frac{\hat{J}^2 - \hat{J}_1^2 - \hat{J}_2^2}{2}\right] |j_1 j_2 j m\rangle$$

$$\lambda \frac{j(j+1) - j_1(j_1+1) - j_2(j_2+1)}{2} |j_1 j_2 j m\rangle .$$

§5 Clebsh-Gordan 系数

- j_1 和 j_2 给定：$(2j_1+1)\times(2j_2+1)$ 维

- 基矢 $\begin{cases} \{|j_1 m_1 j_2 m_2\rangle\}, & \text{无耦合表象} \\ \{|j_1 j_2 j m\rangle\}, & \text{耦合表象} \end{cases}$

- $|j_1 m_1 j_2 m_2\rangle = \mathbb{1}\,|j_1 m_1 j_2 m_2\rangle$

$$= \sum_j \sum_m |j_1 j_2 j m\rangle \underbrace{\langle j_1 j_2 j m | j_1 m_1 j_2 m_2\rangle}_{\text{C-G 系数}}.$$

- $|j_1 j_2 j m\rangle = \mathbb{1}\,|j_1 j_2 j m\rangle$

$$= \sum_{m_1} \sum_{m_2} |j_1 m_1 j_2 m_2\rangle \underbrace{\langle j_1 m_1 j_2 m_2 | j_1 j_2 j m\rangle}_{\text{C-G 系数}}.$$